

Beyond Persistence? PAGE: Persistence-Anchored Gated Forecasting with Spatio-Temporal Graph Networks for Dengue

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Abstract

Dengue causes recurring epidemics in Sri Lanka, and district-level forecasts three weeks ahead give health services time to act. Spatio-temporal graph neural networks (ST-GNNs) and SEIR-informed neural models have been proposed for this task, but both predict the case level directly and are evaluated without the naive persistence forecast, on a benchmark array that we show leaks future data. On this target last week’s count explains $r^2 = 0.89$ of the next and the best climate covariate 0.03, so a direct head must relearn an identity map through its encoder, and several published ST-GNNs fail to. We propose PAGE (Persistence-Anchored Gated Estimation), a forecasting head that plugs into any spatio-temporal encoder and predicts a gated departure from the last observed week, at the cost of one parameter. We correct the benchmark into a leakage-free 559-week dataset, extend it to August 2026 by parsing 127 new reports, and evaluate five published ST-GNNs and the LSTM of SEIR-LSTM on one pre-registered harness. PAGE repairs the three encoders that fail as published by 5.95–16.29 validation RMSE and 23.24–34.50 test RMSE (9/9 paired runs each), and on AAGCN it beats persistence on all 9 runs on both splits. An ablation attributes the gain to the anchor and the gate: an SEIR simulator in the same head adds nothing, and on 85 unseen forecast starts it ties its no-SEIR twin ($\Delta = -0.03$, $p = 0.67$). On those unseen weeks no learned model beats persistence on RMSE (34.75 against 35.43 for PAGE), although PAGE has lower MAE and SMAPE. On this target the useful prior is persistence, not transmission dynamics. Code and data: <https://anonymous.4open.science/r/TODO>.

Keywords

dengue forecasting, spatio-temporal graph neural networks, persistence, SEIR, physics-informed learning, data leakage, prospective evaluation, negative results

1 Introduction

Dengue is the fastest-spreading mosquito-borne viral disease, and Sri Lanka faces recurring severe epidemics: the 2017 outbreak alone recorded over 186,000 cases and 440 deaths [19]. Roughly three weeks separate infection from diagnosis, so district-level forecasts

issued $H = 3$ weeks ahead give clinicians and vector-control teams time to prepare beds and target spraying.

Two families of deep model are proposed for this task. Spatio-temporal graph neural networks (ST-GNNs), transferred from traffic forecasting, treat districts as nodes and learn from neighbours; Weng et al. [21] benchmark five of them on Sri Lanka’s weekly record. SEIR-informed models push the forecast through compartmental dynamics; Liu et al. [11] drive an SEIR model with an LSTM on the same country. The natural hybrid, an SEIR-GNN, replaces that LSTM with a graph encoder.

The technical gap. Weekly dengue incidence is close to a random walk. On the corrected data, cases at $t - 1$ explain $r^2 = 0.89$ of cases at t ; the best single climate covariate at a causal lag explains 0.03. Both model families nevertheless predict the case *level* directly from an encoder representation, so the network must relearn the identity map $\hat{y}_{t+1} \approx y_t$ through its message passing and recurrence before it can add anything. We find that three of the five published ST-GNNs do not: they underfit, and on validation are 5–17 RMSE worse than simply repeating last week. The benchmark that reports them hides this, because it omits the persistence forecast $\hat{y}_{t+h} = y_t$ and its array leaks future data (§3.2). Neither line of work tests its physics or its graph against a matched control that differs in that component alone.

Proposed solution. We propose PAGE (Persistence-Anchored Gated Estimation), a forecasting head that plugs into any spatio-temporal encoder. It starts from the persistence forecast and lets the network supply only a *departure* from it, scaled by a learned gate initialised near zero. The identity map is therefore free, the network learns only what persistence misses, and the gate shrinks that correction until the data support it. PAGE adds one parameter. Its departure can come from a linear read-out or from a differentiable SEIR simulator, which lets one controlled ablation ask whether physics helps.

Main results. On one leakage-free harness (Figure 1, Table 2), PAGE repairs STGAT, A3TGCN and DCRNN by 5.95–16.29 validation RMSE (9/9 paired runs each), and with AAGCN it beats persistence on all 9 runs on validation and on test. The ablation (Table 5) attributes the gain to the anchor and the gate; an SEIR departure adds nothing. On 85 forecast starts from March 2024 to

August 2026, scored with settings frozen before those weeks were parsed, the SEIR branch ties PAGE ($p = 0.67$) and no learned model beats persistence on RMSE, although PAGE has lower MAE and SMAPE (Table 9). We report this negative result as it is.

Contributions.

- (1) **PAGE, a persistence-anchored gated head** (§4), encoder-agnostic and one parameter in size, that makes every published ST-GNN we test usable and isolates, by ablation, where the gain comes from: the anchor and the gate, not the physics.
- (2) **A leakage-free dataset** (§3). We audit the benchmark array, which mixes time orders, trains on post-test data and feeds future rainfall as a “lagged” feature, and correct it into a 559-week (2013–2024) leakage-free series, extended with 127 newly parsed 2024–2026 reports for a prospective test.
- (3) **A controlled and prospective evaluation** (§6): six encoders, matched no-physics and no-graph controls, paired tests, pre-registered decision rules and a prospective 2024–2026 test. It shows that the graph adds 0.06 RMSE, the SEIR branch adds nothing, and the remaining error is informational rather than architectural.

2 Related Work

ST-GNNs for epidemics and dengue. Spatio-temporal graph networks couple a graph operator over regions with a temporal operator over history. The five we evaluate come from traffic forecasting. STGAT and A3TGCN [1] use attention, ASTGCN [5] attends over space and time, AAGCN [18] adapts its adjacency, and DCRNN [10] diffuses over the graph inside a recurrent cell; Graph WaveNet [22] learns the adjacency from node embeddings. For epidemics, Cola-GNN [3] and EpiGNN [23] add cross-location attention and transmission-risk encodings. Weng et al. [21] benchmark the five traffic ST-GNNs on Sri Lanka’s dengue record, and GulMohamed et al. [4] add per-horizon heads, temporal attention and a Gaussian output. All of them predict the case level directly, and the dengue benchmark [21] reports no persistence baseline.

Mechanistic hybrids. Liu et al. [11] forecast Sri Lankan dengue with an SEIR model whose force of infection is predicted by an LSTM; we call this SEIR-LSTM. EINNs [15] couple a neural forecaster with a mechanistic model through shared latent dynamics. CausalGNN [20] and MepoGNN [2] embed a metapopulation SEIR in a graph network, so that districts are coupled inside the dynamics. Physics-informed networks [14] add dynamics as a loss; MP-PINN [12] applies this to epidemic curves. We use the SEIR–SEI stage durations of Phaijoo and Gurung [13]. These works compare against purely data-driven models, but not against the same head with the simulator removed, so the share of the gain that is due to physics is not measured.

Persistence and simple baselines. In classical forecasting the naive forecast $\hat{y}_{t+h} = y_t$ is the reference every method must beat, and differencing, which models the change rather than the level, is the standard treatment of a series close to a random walk [6]. In deep spatio-temporal forecasting, simple models often match complex ones: STID [17] replaces graph operators with identity embeddings. Reversible instance normalisation [8] targets distribution shift, count likelihoods such as the negative binomial of

Table 1: Audit of the benchmark array, recomputed by the reproduction notebook from the original files.

finding	consequence
Rows 0–47 are Dec 2022–Nov 2023; rows 51 onward run 2013–2022	45 rows from 2023 sit in the training split of every fold, which tests on 2018–2022
Cases reordered, weather not	the array’s temperature matches ERA5 at $r = 0.95$ on its own row order and $r = 0.45$ on the dates the case rows carry
“Lag 12” precipitation	matches ERA5 rain 12 weeks <i>later</i> ($r = 0.68$) and not 12 weeks <i>earlier</i> ($r = 0.19$): future information
48 reports dropped, one week stored twice	454 of 459 rows equal a published report exactly
Week-395 spike	row A of one printed table is a dragged formula (15 of 24 cells are the sum of the two before, districts sum to 7,165 against a national cell of 35); the same table’s cumulative row gives the true count

DeepAR [16] target skewed counts, and TimeGAN [24] generates synthetic sequences. PAGE brings the classical idea into the ST-GNN head: it anchors on persistence and learns a gated departure. We also test each of the remedies above under the same protocol.

3 Dataset Construction

3.1 Reproducing the benchmark

We first ran the reference implementation [21] unmodified (their code, array and protocol) on a pinned software stack. It recovers their published numbers to within 7.3% across twenty comparisons. Three properties of their evaluation, however, account for the reported advantage: the “cross validated” column is computed on a loader that includes training data; RMSE is averaged per window rather than pooled, which on this target is ~39% lower; and no naive baseline is reported. Adding persistence to that same column reverses the ranking; it beats *every* model on MAE and RMSE. An independent re-implementation that shares no code with either codebase reproduces this.

3.2 Auditing the benchmark array

The benchmark array (`sri_lanka_2013–2022_shifted.npy`) has no date axis. We dated every row by matching its 25 district counts exactly against the weekly reports, and dated the weather channels separately by finding the calendar offset at which the array’s temperature best matches ERA5. Table 1 lists what that shows.

Two of these are information leaks. Training on 2023 while testing on 2018–2022 lets a model see the future of its test period, and a “lagged” rainfall channel that holds rain from twelve weeks ahead gives it that future’s weather. Any score on this array, including the published ones and our own earlier work, is therefore an optimistic bound. Earlier drafts of this study called the week-395 spike an administrative reporting backlog. It is a spreadsheet error, and it can be corrected from the same published table.

3.3 The corrected dataset

The 2013–2024 case counts come from the benchmark authors’ table of parsed Table 1 rows of every Weekly Epidemiological Report; we do not re-parse those reports. The other inputs we obtain ourselves: the one report PDF used for the correction, ERA5 daily responses from the Open-Meteo archive, and the Department of Census and Statistics population and census PDFs. Every input file has a recorded SHA-256 checksum. From these the case series is re-dated, keyed by report volume and number: 559 weeks from 2013-W26 to 2024-W10. The seven weeks with no report stay missing and are never interpolated; districts plus the Kalmunai division equal the Epidemiology Unit’s own national total in 432 of 552 reports. Weekly ERA5 temperature, precipitation, humidity and soil moisture are aggregated from daily values at an interior point of each district, and each week’s population is the latest official figure for an earlier year (the 2012 Census, whose 25 district totals sum exactly to the national total, then mid-year estimates).

Lags are applied in the model, not baked into the data: a forecast for week t reads cases up to $t - 1$ and climate up to $t - 2$, since ERA5 is published about five days late. The lags are enforced in code, and for the data loader used in the prospective test a perturbation test asserts that no value after either limit can change a forecast. On this series the persistence test RMSE on the three development origins (§5.3) is **36.02**, against 44.80 on the array (29.52 with the week-395 windows excluded). The last test fold covers the 2023 epidemic.

3.4 Extension to 2026 for a prospective test

To test on weeks that no design decision had seen, we extend the series with the 127 Weekly Epidemiological Reports from 2024-W11 to 2026-W32. The parsing and quality rules (district totals must match the national total; a dragged-formula pattern or a reprinted table voids a week’s row, which is then replaced by the difference of the cumulative rows or left missing) were committed before any count was read. The result is 109 observed weeks, 8 corrected and 10 missing; nothing is interpolated. One deviation is reported in full: a first scoring run exposed two reconstruction defects (the first report of a year carries the previous year’s total in its cumulative row, and two reports reprint the previous week’s table). The rules were amended on report semantics alone, no model setting changed, and only the re-parsed data are scored (§6.5). This is therefore an amended prospective evaluation, not an untouched one.

4 The PAGE Framework

4.1 Problem formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ be the district graph with $|\mathcal{V}| = N$ districts and adjacency $\mathbf{A} \in \{0, 1\}^{N \times N}$ (with self-loops), and let $y_{i,t} \geq 0$ be the dengue cases reported by district i in week t . Counts are modelled on the scale $z_{i,t} = (\log(1 + y_{i,t}) - m)/s$, where m and s are the mean and standard deviation of $\log(1 + y)$ over the training weeks only. At forecast origin t the input is the window $\mathbf{Z}_t = [z_{i,t-W+1}, \dots, z_{i,t}]_{i \in \mathcal{V}} \in \mathbb{R}^{N \times W}$ and, optionally, exogenous covariates $\mathbf{E}_t \in \mathbb{R}^{N \times F}$ observed at causal lags (§3.3). The task is to predict $\mathbf{Y}_t = [y_{i,t+h}] \in \mathbb{R}^{N \times H}$ for horizons $h = 1, \dots, H$.

4.2 Encoder

Any spatio-temporal encoder f_θ maps the window to one representation per district,

$$\mathbf{H} = f_\theta(\mathbf{Z}_t, \mathbf{A}) \in \mathbb{R}^{N \times d}, \quad \tilde{\mathbf{h}}_i = [\mathbf{h}_i \parallel \mathbf{e}_{i,t}], \quad (1)$$

where $\parallel$ concatenates any covariates around the encoder, so that a published architecture runs exactly as designed. We use the five published ST-GNNs and the LSTM of SEIR-LSTM unchanged (§5.2). A linear read-out gives a level forecast $\mathbf{u}_i = \mathbf{W}\tilde{\mathbf{h}}_i + \mathbf{b} \in \mathbb{R}^H$; the published models stop here and output $\hat{z}_i = \mathbf{u}_i$ (the *direct* head).

4.3 Persistence-anchored gated head

PAGE anchors every horizon on the last observed week, $p_i = z_{i,t}$, and lets the network contribute only a gated departure from it:

$$\delta_i = \mathbf{u}_i - p_i \mathbf{1}_H, \quad \hat{z}_i = p_i \mathbf{1}_H + g \delta_i, \quad g = \sigma(a), \quad (2)$$

with one learned scalar a , initialised at $a_0 < 0$ so that training starts close to the persistence forecast. Equivalently $\hat{z}_i = (1 - g) p_i \mathbf{1}_H + g \mathbf{u}_i$: a learned convex combination of persistence and the encoder’s forecast. Counts are recovered as $\hat{y}_{i,t+h} = \max(0, \exp(s \hat{z}_{i,h} + m) - 1)$.

Design rationale. Three properties follow from Eq. (2). (i) *The identity is free.* At $g \rightarrow 0$ the model is exactly persistence, which already explains $r^2 = 0.89$ of the next week; a direct head must instead carry the level through the encoder’s message passing and recurrence, and the three failing encoders do not (§6.1). (ii) *The gate is a learned shrinkage.* The gradient raises g only as far as $\mathbf{u}_i$ improves on p_i , so a weak encoder cannot pull the forecast far from a strong baseline. (iii) *It costs one parameter* and no extra computation (Table 8). Removing the gate ($g \equiv 1, \hat{z}_i = p_i \mathbf{1}_H + \delta_i$ with an unconstrained departure) gives the *residual* head, and removing the anchor gives the direct head; both are ablations in §6.2.

4.4 Optional SEIR branch

To test whether transmission dynamics help, the read-out can be replaced by a differentiable SEIR model. The encoder predicts a weekly force of infection $\lambda_{i,h} = \min(\exp(\mathbf{w}^\top \tilde{\mathbf{h}}_i + b_\lambda), \lambda_{\max})$, and the compartments $\dot{S} = -\lambda S$, $\dot{E} = \lambda S - \omega E$, $\dot{I} = \omega E - \gamma I$, $\dot{R} = \gamma I$ are stepped daily from a state $\mathbf{s}_{i,t}$ seeded by observed cases (exposed and infectious from the counts of weeks $t - 1$ and t , divided by a reporting rate ρ ; susceptibles from a serosurvey). Weekly incidence $\iota_{i,h}$ (the flow $E \rightarrow I$) times ρ times the district population P_i gives counts, so $u_{i,h} = (\log(1 + \rho P_i \iota_{i,h}) - m)/s$, which enters Eq. (2) in place of the read-out; learned scales on the seeded exposed state and on ρ are allowed. With the anchor and gate this is the SEIR-GNN (or SEIR-LSTM when f_θ is the LSTM); without them, $\hat{z}_i = \mathbf{u}_i$, it is a pure SEIR decoder.

4.5 Learning objective

All parameters $\Theta = \{\theta, \mathbf{W}, \mathbf{b}, a\}$ (plus $\mathbf{w}, b_\lambda$ and the SEIR scales when the branch is used) minimise the squared error on the standardised scale over a mini-batch $\mathcal{B}$ of forecast origins,

$$\mathcal{L}(\Theta) = \frac{1}{|\mathcal{B}|NH} \sum_{t \in \mathcal{B}} \sum_{i=1}^N \sum_{h=1}^H (\hat{z}_{i,h}^{(t)} - z_{i,t+h})^2, \quad (3)$$

with early stopping on validation RMSE in counts. We also test a negative-binomial (NB) objective, which adds a dispersion head

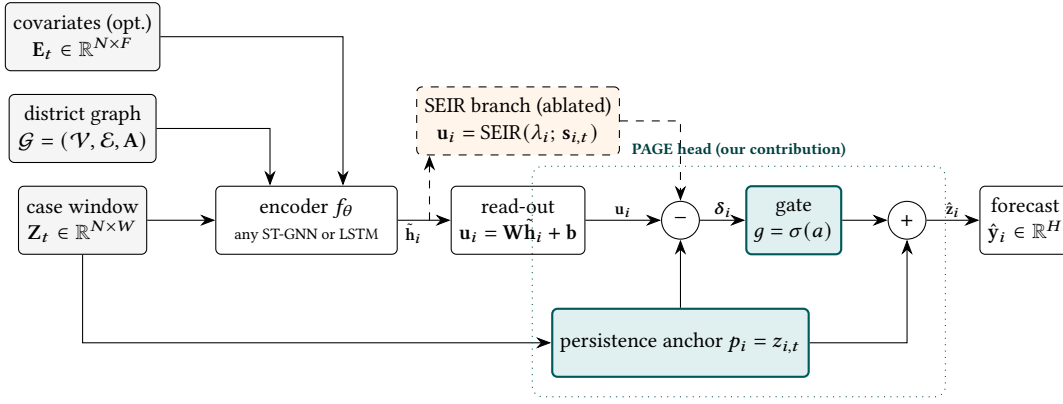


Figure 1: PAGE. An unchanged encoder reads the last W weeks of standardised log cases on the district graph. Its read-out u_i is turned into a departure $\delta_i = u_i - p_i$ from the persistence anchor, shrunk by a learned gate, and added back to the anchor. The dashed SEIR branch replaces the read-out with a differentiable simulator; it is the component the ablation removes.

and minimises the NB2 negative log-likelihood with mean $\mu = \exp(s\hat{z} + m) - 1$. The reason is a bias: under squared error on $\log(1+y)$, exp of the fit is a conditional *median*, while RMSE rewards the conditional *mean*, which NB’s μ estimates.

4.6 Implementation details

$N = 25$ districts, window $W = 3$, horizon $H = 3$, hidden size $d = 64$, gate initialisation $a_0 = -2$ ($g_0 = 0.12$). The graph is the benchmark’s district adjacency with self-loops (141 directed entries; 139 in the prospective test, after two non-border entries are removed), row-normalised; AAGCN runs with its adaptive adjacency off, as published. SEIR rates are per day: incubation $\omega = 0.1$, recovery $\gamma = 1/7$, reporting $\rho = 1/11$, $\lambda_{\max} = 1/7$, initial susceptible fraction $1 - 0.682$ from a 2013–14 Colombo serosurvey [7], with stage durations of [13]. Training uses Adam (learning rate 3×10^{-3} with cosine decay, weight decay 10^{-4}), batch size 32, dropout 0.1, at most 300 epochs (400 for NB) with early stopping on validation RMSE, patience 40, and the best-validation weights restored. Seeds are 0, 1, 2. All models are written in plain PyTorch with dense $N \times N$ graph operations and no graph library; each published encoder is checked against its reference implementation in unit tests. The main study (810 runs) ran in 3.0 hours on a 4-core Kaggle CPU, each run single-threaded; the prospective study ran on a laptop CPU (Intel Core i5-12450H). No GPU is needed.

5 Experimental Setup

5.1 Dataset overview

The corrected series (§3.3) covers 25 districts over 559 weeks, 2013-W26 to 2024-W10, with weekly cases, ERA5 temperature, precipitation, humidity and soil moisture, population, and the benchmark’s district graph. Seven weeks are missing and windows that touch them are dropped. The extension (§3.4) adds 127 weeks to 2026-W32 for the prospective test only. Every model, baseline and control is scored on exactly the same validation and test windows and trained on the same training windows, except that arms with climate lags skip the first training weeks, which have no climate history that far back.

5.2 Baselines

Following the brief’s baseline rule (LSTM for time series, GCN for graphs), and the literature on this dataset, we compare against:

- **Persistence** $\hat{y}_{t+h} = y_t$ and **seasonal naive** $\hat{y}_{t+h} = y_{t+h-52}$, the classical floors.
- **LSTM**, the two-layer encoder of SEIR-LSTM [11], and a **GCN** [9] against the same network without message passing, the mandatory sequence and graph baselines.
- The **five published ST-GNNs** of the Sri Lankan benchmark [21]: STGAT, A3TGCN, ASTGCN, AAGCN and DCRNN, each with its direct head as published.
- **SEIR-informed models**: the SEIR-LSTM [11], its graph version (SEIR-GNN), a pure SEIR decoder, and a metapopulation SEIR head in the style of MepoGNN [2] and CausalGNN [20].
- **Classical and simple learners**: histogram gradient-boosted trees (scikit-learn) with and without climate, a negative-binomial GLM, k -nearest-neighbour analogues, an AR(3) ridge regression, and an STID-style MLP [17].

5.3 Evaluation protocol

Splits. Rolling origins at 0.55, 0.70 and 0.85 of the series, three seeds each. The 30 windows before each origin are validation, the following 15% are test, and normalisation uses training weeks only; no window is shuffled across time. A confirmation uses nine disjoint origins at $0.40 + k/15$ ($k = 0, \dots, 8$). The prospective test trains once on all development weeks and scores the 85 forecast starts from 2024-W11 onward, with no target week shared between partitions.

Metrics. RMSE in weekly cases, pooled over every (window, district, horizon) cell, is primary; MAE, SMAPE and MAPE (on targets ≥ 1) are secondary, overall and per horizon. Tables report the mean over runs.

Selection and tests. Selection uses validation RMSE only; test numbers are written to disk and never used to choose. The protocol and, for each experiment, its arms, control, decision rule and predicted outcome were committed before it ran. An arm is adopted if its mean validation difference to its control is negative, it wins at

Table 2: Comparative analysis on one harness (corrected data, 3 origins $\times$ 3 seeds). Mean validation / test RMSE in weekly cases. Top: the five published ST-GNNs and the LSTM of SEIR-LSTM, as published (direct head), with PAGE, and with PAGE plus the SEIR branch (the SEIR-GNN, or SEIR-LSTM for the LSTM). Squared error, no seasonal features. Bottom: baselines; B is the configuration validation selects. Selection uses validation only.

encoder	direct	PAGE	PAGE+SEIR
AAGCN	16.71 / 35.15	16.57 / 33.53	17.43 / 35.58
ASTGCN	16.83 / 35.61	16.76 / 34.76	17.51 / 35.71
LSTM	16.91 / 35.10	16.94 / 35.85	17.69 / 35.52
STGAT	22.93 / 62.00	16.98 / 36.46	17.63 / 35.84
A3TGCN	28.63 / 60.40	17.68 / 37.16	17.68 / 35.50
DCRNN	34.64 / 73.52	18.35 / 39.02	17.67 / 35.52
baseline	validation / test		
Persistence	17.86 / 36.02		
No message passing (res.)	16.90 / 34.57		
GCN (res.)	16.84 / 34.50		
STID-style MLP [17]	16.35 / 36.33		
k -NN analogues	16.52 / 34.34		
NB-GLM	16.97 / 42.85		
Boosted trees	17.94 / 38.96		
Boosted trees + climate	17.40 / 36.49		
B : AAGCN, NB, season	15.51 / 37.54		

least 7 of 9 (origin, seed) units, and it does not raise horizon-3 error. Differences are paired on matched (origin, seed) units and tested with an exact two-sided sign-flip permutation test, BH-adjusted within each family. With three origins this unit mostly measures run-to-run stability; claims about the series need the nine-origin protocol, where the origin-level test can reach $p = 0.004$. The prospective test uses a block bootstrap over forecast starts (block 8, 10,000 replicates) with Holm correction over its two pre-registered comparisons.

6 Experiments and Results

All numbers are RMSE in weekly cases on the corrected dataset. Sections 6.1–6.4 and 6.6–6.8 come from the 810 runs of the reproduction notebook; §6.5 comes from the prospective study. Validation decides and test is reported beside it.

6.1 Comparative analysis

Table 2 compares every encoder and baseline on identical windows, folds, loss, early stopping and metric, and Table 6 pairs PAGE against the published head and against persistence.

As published, three of five ST-GNNs fail. Behind the direct head AAGCN, ASTGCN and the LSTM score 16.71, 16.83 and 16.91 on validation, a spread of 0.20, and all beat persistence (17.86). STGAT, A3TGCN and DCRNN score 22.93, 28.63 and 34.64, worse than persistence by 5–17. They are underfitting rather than short of information: in their NB, seasonal-feature configuration, a linear model fitted to their training residuals recovers 48% and 60% of STGAT’s and A3TGCN’s validation residual variance, against 3–7% for the working encoders.

Table 3: Every lever, paired against its own control on matched (origin, seed) runs. Δ is validation RMSE (negative is better); wins count the 9 units where the arm is better; p_{adj} is BH across this table. Test Δ is shown and decided nothing. With three origins these p -values describe run-to-run stability, not the series (§5.3).

lever	Δ val	wins	p_{adj}	Δ test
<i>Worked</i>				
NB likelihood (vs squared error)	-0.76	9/9	0.014	+1.61
Seasonal features	-0.76	6/9	0.185	+0.70
Best model B vs persistence	-2.35	9/9	0.014	+1.53
<i>Graph and architecture</i>				
Graph (GCN vs no message passing)	-0.06	3/9	0.501	-0.07
Learned adjacency	-0.14	5/9	0.185	-0.33
District seasonal curves	-0.05	6/9	0.757	+0.11
Nonlinear (MLP) head	+0.28	1/9	0.027	-0.31
MLP head + climate lags 2–13	+0.13	3/9	0.482	-0.70
Global context node	+0.02	3/9	0.844	-1.40
Residual head + NB	+0.47	2/9	0.207	-2.96
Combined architecture changes	+0.55	2/9	0.078	+0.59
<i>Literature remedies</i>				
RevIN	+1.47	0/9	0.014	-0.42
District identity embedding	+0.19	3/9	0.219	+0.98
STID-style MLP	+0.84	0/9	0.014	-1.22
NB-GLM (no network)	+1.46	0/9	0.014	+5.31
k -NN analogues (no network)	+1.01	0/9	0.014	-3.20
Ensemble B + NB-GLM + k -NN	+0.06	4/9	0.501	-1.79
<i>Data</i>				
Half the training windows	+0.39	2/9	0.050	+0.47
TimeGAN augmentation	+0.60	0/9	0.014	+0.52
SEIR-simulated pre-training	+0.03	6/9	0.822	+0.92
Climate, lags 2–4	+0.18	2/9	0.199	+0.51
Climate, lags 2–13	+0.19	2/9	0.302	-0.05
Climate, lags 2–25	+0.47	0/9	0.014	-0.47
<i>Physics</i>				
SEIR as auxiliary loss	+0.02	4/9	0.812	-0.32
SEIR simulator vs no-physics twin (AAGCN)	+0.86	1/9	0.020	+2.05
Spatial physics penalty	-0.06	6/9	0.207	-0.02
Metapopulation vs gated SEIR head	+0.20	1/9	0.020	+0.04
Gated SEIR-GNN vs SEIR-LSTM	-0.14	8/9	0.020	+0.02

PAGE makes every encoder usable. With PAGE all six encoders score 16.57–18.35 on validation and 33.53–39.02 on test. It repairs the three failing encoders by 5.95, 10.95 and 16.29 on validation and 25.54, 23.24 and 34.50 on test, on 9/9 paired runs each ($p_{\text{adj}} = 0.004$). On the working encoders it changes validation by at most 0.14 and improves AAGCN’s test RMSE on 9/9 runs (−1.62).

Against persistence, the strong encoders win. AAGCN with PAGE beats persistence on all 9 runs on validation (−1.29) and on test (−2.48), and ASTGCN on 9/9 and 8/9 (−1.10, −1.26). The repaired encoders reach persistence but do not go clearly beyond it, and DCRNN with PAGE stays slightly worse (+0.50). The anchor removes failure; it cannot add information the encoder does not have.

Baselines. The graph adds little: a GCN beats the same network without message passing by 0.06 (3/9, Table 3). k -NN analogues, with no network, reach 16.52 / 34.34; gradient-boosted trees reach

Table 4: The best model and the SEIR-GNN against their controls. Three-origin rows pair 9 (origin, seed) runs; nine-origin rows pair the 9 disjoint origins (seeds averaged), the unit a claim about the series rests on (attainable p floor 0.004). Raw two-sided p .

comparison	validation			test		
	Δ	wins	p	Δ	wins	p
B vs persistence, 3 origins	-2.35	9/9	0.004	+1.53	3/9	0.230
B vs persistence, 9 origins	-2.58	8/9	0.105	+2.24	4/9	0.293
gated SEIR-GNN (ASTGCN) vs SEIR-LSTM, 3 origins	-0.54	9/9	0.004	+0.14	5/9	0.594
gated SEIR-GNN (AAGCN) vs SEIR-LSTM, 3 origins	-0.14	8/9	0.008	+0.02	5/9	0.855
AAGCN vs LSTM, direct head, 3 origins	-0.27	6/9	0.082	+1.96	1/9	0.012
gated SEIR-GNN (ASTGCN) vs SEIR-LSTM, 9 origins	-0.62	8/9	0.012	+0.07	6/9	0.883
gated SEIR-GNN (AAGCN) vs SEIR-LSTM, 9 origins	+0.18	6/9	0.797	-0.30	5/9	0.078
metapopulation SEIR-GNN vs SEIR-LSTM, 9 origins	+0.49	4/9	0.398	-0.01	4/9	0.961
AAGCN vs LSTM, direct head, 9 origins	-0.72	6/9	0.516	-0.88	5/9	0.820
gated SEIR-GNN (AAGCN) vs persistence, 9 origins	-1.98	8/9	0.137	-0.59	6/9	0.680

Table 5: Ablation: from the published direct head to PAGE. Each encoder behind four heads that add, in turn, the persistence anchor (residual), the learned gate (PAGE) and the SEIR branch (mean validation RMSE; configuration of Table 2). The last two columns pair PAGE+SEIR with PAGE; negative favours the SEIR branch, and the count is runs where it wins.

encoder	direct	resid.	PAGE		SEIR branch	
			PAGE	+SEIR	val	test
AAGCN	16.71	16.72	16.57	17.43	+0.86 (1/9)	+2.05 (0/9)
ASTGCN	16.83	16.85	16.76	17.51	+0.75 (0/9)	+0.95 (3/9)
LSTM	16.91	16.96	16.94	17.69	+0.75 (0/9)	-0.33 (6/9)
STGAT	22.93	17.85	16.98	17.63	+0.66 (1/9)	-0.62 (8/9)
A3TGCN	28.63	17.47	17.68	17.68	-0.00 (6/9)	-1.66 (9/9)
DCRNN	34.64	17.52	18.35	17.67	-0.68 (3/9)	-3.50 (9/9)
persistence		17.86				

Table 6: PAGE against the published direct head and against persistence, on matched (origin, seed) runs of Table 2. Mean RMSE difference (negative favours PAGE) and wins out of 9. *: $p_{\text{adj}} < 0.05$, exact sign-flip test, BH across the table (attainable floor 0.004).

encoder	vs direct (published)		vs persistence	
	val	test	val	test
AAGCN	-0.14 6/9	-1.62 9/9*	-1.29 9/9*	-2.48 9/9*
ASTGCN	-0.07 7/9	-0.85 6/9	-1.10 9/9*	-1.26 8/9*
LSTM	+0.03 3/9	+0.75 2/9	-0.92 9/9*	-0.17 3/9
STGAT	-5.95 9/9*	-25.54 9/9*	-0.88 6/9	+0.44 3/9
A3TGCN	-10.95 9/9*	-23.24 9/9*	-0.17 3/9	+1.14 3/9
DCRNN	-16.29 9/9*	-34.50 9/9*	+0.50 3/9	+3.00 3/9

17.94, or 17.40 with climate. The configuration selected on validation under the adoption rule, B (AAGCN, direct head, NB loss, seasonal features; 15.51), is worse than persistence on test (37.54). AAGCN with PAGE has the lowest test RMSE of every configuration trained on the three origins (33.53), but validation would not have chosen it, so we report it as observed rather than selected.

Table 7: Hyperparameter tuning: mean validation RMSE over the nine purged development origins $\times$ 3 seeds (108 runs per arm). Bold: the setting each arm selects. The same grid and budget are given to every learned arm; test data play no part. AR(3) ridge, by α : 0.01: 27.64, 0.1: 27.64, 1: 27.64, 10: 27.65, 100: 27.76.

learning rate	hidden d	PAGE	PAGE+SEIR	residual
1×10^{-3}	32	25.39	25.35	25.38
1×10^{-3}	64	25.19	25.36	25.01
3×10^{-3}	32	25.19	25.10	24.98
3×10^{-3}	64	24.90	24.92	24.81

Table 8: Computational cost. Parameters of the full model behind the direct head, and the parameters each head adds. Seconds per epoch and seconds per training run (with early stopping) are means over the 9 runs of the Kaggle run, each run single-threaded on a 4-core Kaggle CPU. PAGE adds one parameter and no measurable time; the SEIR branch adds five parameters and up to $2.2\times$ the time per epoch (AAGCN).

encoder	parameters			s / epoch			s / run
	direct	PAGE	+SEIR	direct	PAGE	+SEIR	PAGE
AAGCN	2,873	+1	+5	0.19	0.20	0.41	19
ASTGCN	50,769	+1	+5	0.92	0.97	1.16	62
LSTM	50,627	+1	+5	0.46	0.49	0.68	35
STGAT	160,894	+1	+5	0.23	0.26	0.42	14
A3TGCN	8,742	+1	+5	0.23	0.23	0.36	14
DCRNN	827,843	+1	+5	4.08	3.85	3.56	220

6.2 Ablation study

Table 5 builds PAGE from the published head one component at a time: the persistence anchor (residual head), then the gate (PAGE), then the SEIR branch. A pre-registered rule credits physics only if the SEIR branch beats PAGE on at least 7 of 9 runs with $p_{\text{adj}} < 0.05$ on each failing encoder.

The anchor removes the failure. A residual head, with no gate, already brings the failing encoders to 17.47–17.85. **The gate**

Table 9: Prospective evaluation on weeks no model or setting had seen: 85 forecast starts, 2024-W11 to 2026-W32. Settings frozen on development data before the new weeks were parsed. Neural arms: mean over 3 seeds, seed range of RMSE in brackets. Bold: best per metric. Pre-registered tests (block bootstrap over forecast starts, Holm over two): P1, SEIR branch vs PAGE: $-0.03 [-0.41, +0.12]$, $p_{\text{Holm}} = 0.67$; P2, PAGE+SEIR vs persistence: $+0.65 [-0.99, +1.69]$, $p_{\text{Holm}} = 0.62$.

arm	RMSE	MAE	SMAPE	MAPE	RMSE $h=1/2/3$
Persistence	34.75	11.78	50.3	53.3	15.4/35.7/46.0
PAGE (adaptive graph)	35.43 [35.24, 35.60]	11.35	45.7	43.9	16.0/35.7/47.3
PAGE + SEIR branch	35.40 [35.11, 35.56]	11.29	46.1	43.5	15.8/35.7/47.3
Residual head (no gate)	35.74 [35.37, 36.19]	11.61	46.6	43.0	16.4/36.0/47.6
AR(3) ridge	36.35	11.35	46.7	44.5	17.3/35.9/48.8
Seasonal naive	62.43	27.53	74.6	113.4	50.7/62.5/72.2

improves on it for STGAT (17.85 to 16.98), AAGCN and ASTGCN, and costs A3TGCN 0.21 and DCRNN 0.83; on the prospective weeks it beats the ungated head ($-0.31 [-0.77, -0.08]$, Table 9). Anchor and gate together account for 96–112% of the repair that the full SEIR head achieves on each failing encoder.

The SEIR branch adds nothing. Added to PAGE, the simulator earns credit on none of the three failing encoders: it is significantly worse on STGAT ($+0.66$, 1/9, $p_{\text{adj}} = 0.023$), ties on A3TGCN (-0.00 , 6/9), and helps DCRNN on only 3/9 runs. On the working encoders it costs 0.75–0.86 (0 or 1 of 9 runs). On test the sign reverses for the failing encoders (up to -3.50 , 9/9); this was not used to decide and is the validation–test disagreement of §6.8.

Why the pure SEIR decoder fails. Routing the forecast entirely through the simulator scores 24.01–25.75, worse than persistence on every encoder and 7.3–7.4 worse than the direct head on each working one. Inverting the simulator for the force of infection that reproduces each observed count shows why: with transmission off, the exposed state seeded from recent cases still emits *more* cases than observed for 14–16% of training targets, which no network output can reach; the required force of infection is only $r^2 = 0.22$ –0.26 predictable from case history, against 0.81–0.82 for the direct target; and susceptible depletion is not the cause (0.7–4.9% of cells reach the clamp).

Loss and other levers. Table 3 and Figure 2 pair every further lever against its own control. The NB likelihood is the largest controlled validation gain (-0.76 , 9/9, $p_{\text{adj}} = 0.014$), larger than any architecture change, because it corrects the median–mean bias of §4.5; on test it is 1.61 worse, so we do not adopt it in PAGE. No architecture change improves validation by more than 0.15, and the literature’s remedies are neutral or worse: RevIN $+1.47$ (0/9), an STID-style MLP $+0.84$ (0/9), TimeGAN augmentation $+0.60$ (0/9), climate at lags 2–25 $+0.47$ (0/9).

6.3 Hyperparameter tuning

In the main study every encoder and head uses the same fixed settings (§4.6), chosen before the runs, so no model gets more tuning than another. Before the prospective test we tuned each learned arm on the same grid and budget: learning rate $\{10^{-3}, 3 \times 10^{-3}\} \times$ hidden size $\{32, 64\}$, 108 runs per arm on the nine development origins, selected on validation only. All three arms select learning rate 3×10^{-3} and $d = 64$, the settings of the main study (Table 7). The whole grid moves validation RMSE by at most 0.57, and the ridge penalty of AR(3) by 0.12, so the conclusions do not hinge on

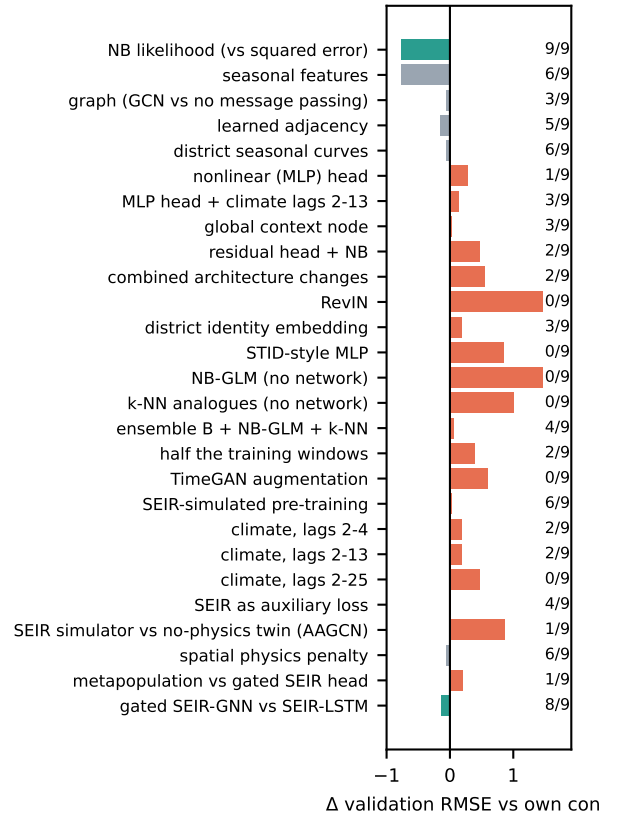


Figure 2: Every lever’s validation difference against its own control (Table 3); win counts on the right. Green: negative and $\geq 7/9$ wins. Grey: negative but inconsistent. Red: worse.

tuning. Halving the training windows costs B 0.39 and a quarter of them 0.25 (Table 3); the models are not short of data.

6.4 Computational analysis

Table 8 gives the cost of each encoder and head. PAGE adds one parameter and no measurable time per epoch (AAGCN 0.19 to 0.20 s). The SEIR branch adds five parameters but up to 2.2× the time

per epoch (AAGCN 0.19 to 0.41 s), because it steps the simulator daily within every forecast week. The encoders span three orders of magnitude, from AAGCN (2,873 parameters, 19 s per run) to DCRNN (827,843 parameters, 3.6–4.1 s per epoch), and the smallest is the most accurate. Every model trains on a free CPU; the full 810-run study takes 3.0 hours.

6.5 Prospective evaluation, 2024–2026

The real test is weeks no setting was chosen on. We froze six arms and their settings on development data, then parsed the new weeks (§3.4) and scored each arm once on 85 forecast starts. The neural arms share one encoder, a Graph WaveNet-style network with a learned adjacency [22], the development finalist: PAGE, PAGE with the SEIR branch (the pre-registered proposed model), and the residual head. Persistence, seasonal naive and an AR(3) ridge regression complete the list.

Table 9 gives the result. **No learned model beats persistence on RMSE:** persistence scores 34.75, PAGE 35.43 and PAGE+SEIR 35.40, differences that are not significant (PAGE+SEIR +0.65, $p_{\text{Holm}} = 0.62$). **The SEIR branch adds nothing** over PAGE (−0.03 [−0.41, +0.12], $p_{\text{Holm}} = 0.67$). On the secondary metrics the learned arms do better than persistence (MAE 11.35 against 11.78, SMAPE 45.7 against 50.3), and every arm beats the seasonal naive forecast by 26.7–27.7 RMSE. RMSE is dominated by the 2026 outbreak: persistence scores 14.72, 20.05 and 160.51 in 2024, 2025 and 2026, and PAGE 14.32, 21.43 and 162.75. PAGE is better in the quiet year and worse in the outbreak, where the next large move is not in the inputs. The development advantage of the SEIR arm over persistence (27.34 against 28.54) does not carry over to later data.

Because the SEIR branch ties its no-SEIR twin, we present the simpler PAGE as the framework. That choice is made after seeing the prospective result, and we state it as such: the pre-registered finalist was the SEIR variant, and the two are equivalent.

6.6 SEIR-GNN against SEIR-LSTM, and physics as structure

Behind the same gated SEIR head the graph beats the LSTM on three origins on validation (ASTGCN −0.54, 9/9; AAGCN −0.14, 8/9), but on nine disjoint origins the advantage is not robust to the encoder or the metric (Table 4): ASTGCN’s holds on validation (−0.62, 8/9, $p = 0.012$) and vanishes on test, and AAGCN’s reverses. Two further physics structures do not help: a spatial penalty on relative incidence across borders (−0.06, 6/9) and a metapopulation SEIR head that couples districts inside the dynamics (+0.20 against the gated SEIR head, 1/9, $p_{\text{adj}} = 0.020$). The data explain why: on the district graph, Moran’s I is 0.29 for log cases but −0.03 for weekly log growth, so levels cluster in space while changes in transmission do not.

6.7 The ceiling is informational

The working encoders, the SEIR-LSTM and k -NN, which has no network and no training, leave residuals that correlate 0.91–0.97 with B ’s: the models make the same errors. The remaining error is a lag on moves. B under-predicts rising weeks by 11.5 cases and over-predicts falling weeks by 10.1, and the top 5% of district-weeks

carry 54–67% of every model’s squared error. That is what a mean-seeking objective produces on a near-random-walk target whose next move is not in the inputs.

6.8 Validation and test disagree, systematically

Across the eight families of runs, the configuration validation selects is worse than persistence on test in five, and the best configuration on test is anchored to the last observation in six (PAGE, residual, gated SEIR or k -NN). A 30-week validation span immediately before each origin appears to reward fitting that span’s level, which does not carry into the next period. This is one reason we report the prospective test as the decisive evidence.

7 Discussion

Why PAGE works, and what it cannot do. Weekly dengue counts are close to a random walk, so the most informative feature is the last value. A direct head must carry that value through the encoder; STGAT, A3TGCN and DCRNN lose it and underfit. PAGE hands the level to the output, so the encoder learns only the departure, and the gate keeps a weak departure from pulling the forecast away from persistence. This is why the same head repairs three different architectures, attention, recurrent and diffusion based, by 6–16 RMSE. It also explains its limit: the anchor removes failure, but it cannot create information. When the next large move is not in the inputs, as in the 2026 outbreak, every model, PAGE included, lags it, and on RMSE no learned model beats persistence on later data.

Where physics helps a graph network, and where it does not. On this target the SEIR model cannot be the forecaster: a pure SEIR decoder inherits the poor predictability of its force of infection, and 14–16% of targets are out of its reach. Inside the gated head it looked useful, because it made every encoder usable, but the ablation shows that the anchor and gate do that work, and on later data the simulator ties its no-physics twin. Coupling districts inside the dynamics does not help either, because transmission changes have almost no local spatial structure at district-week resolution. The graph itself adds 0.06 RMSE. The prior that matters is the one the data already state: next week looks like this week.

What the evaluation taught us. Each of the following would have produced a publishable positive result had it gone unchecked: the benchmark array leaks future data through its row order and its climate channels; our own first SEIR-GNN pipeline compared one origin against a three-origin baseline; the gated SEIR head’s repair looked like physics and was the anchor; the validation-selected model B is worse than persistence on test; and the SEIR arm’s development advantage over persistence disappears on 2024–2026 data. We therefore recommend that dengue forecasting papers report persistence, a matched no-physics control and, where possible, a prospective test.

Limitations. One country and one disease at district-week resolution. Three origins cannot support claims at the origin unit, and the nine-origin test spans were examined in earlier experiments, so they are not fresh. The prospective test is amended (§3.4) and uses one encoder family; it is a single 2.4-year period dominated by one outbreak year. The graph is the benchmark’s hand-built adjacency, and mobility data might couple districts differently. The

SEIR branch uses fixed stage durations and one initial susceptible fraction for every district.

8 Conclusion

We proposed PAGE, a persistence-anchored gated head for spatio-temporal dengue forecasting that plugs into any encoder at the cost of one parameter, together with a corrected, leakage-free dataset extended to 2026 with newly parsed reports. PAGE turns three published ST-GNNs that are worse than persistence into usable forecasters, and with a strong encoder it beats persistence on every paired run in the development study. An ablation shows that the gain comes from the anchor and the gate; an SEIR simulator in the same head adds nothing, on development data or on later weeks. On weeks from 2024 to 2026 no learned model beats persistence on RMSE, though PAGE is better on MAE and SMAPE. For district-level dengue in Sri Lanka, the useful prior is persistence, not transmission dynamics, and the next gains must come from inputs that anticipate outbreaks rather than from the architecture. The dataset, the notebook that produces every development number, the pre-registrations and the experiment log are available at <https://anonymous.4open.science/r/TODO>.

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